

From Green-hyperbolic Complexes to Conformal Detours

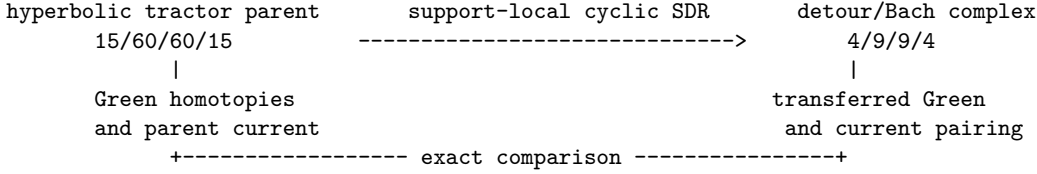
Adjacency bridge note

Bridge note for Lorentzian PDE, Green-hyperbolic-complex, and BGG/detour geometry researchers. Status: theorem-extraction draft, 17 July 2026.

1. Their object and the unresolved question

Benini, Musante, and Schenkel define a Green-hyperbolic complex through retarded and advanced Green homotopies, prove uniqueness up to a contractible space of choices, identify the causal quasi-isomorphism, and compare covariant and fixed-time Poisson structures up to homotopy ([arXiv:2207.04069](#)). Gover, Somberg, and Souček construct formally self-adjoint Yang–Mills detour complexes and obtain conformal BGG-related complexes from tractor connections on Bach-flat four-manifolds ([arXiv:math/0606401](#)). Neither scope is being challenged here.

Our question is constructive: if a hyperbolic tractor/prolonged parent is reduced to a fourth-order detour complex by differential contractions, homological perturbation, and finite support-local shears, what hypotheses ensure that retarded/advanced homotopies, cyclic adjoints, and current pairings descend together?



2. Exact dictionary

Item	Green-hyperbolic/detour language	This project	Conversion or caveat
Differential	cochain differential	classical BV–BFV differential	Degree and sign conventions are fixed by the odd pairing.
Propagation	retarded/advanced Green homotopy	q_1 $\Lambda_{\pm}$ on compact or spacelike-compact support	These are chain homotopies, not an inverse of the isolated Bach operator.
Reduction	quasi-isomorphism or contraction	cyclic differential SDR plus curved HPL and shears	Every shear is finite order and support-local.
Geometric parent	bundle complex with connection	adjoint-tractor/Yang–Mills parent and curvature prolongation	The concrete theorem uses conformal flatness of the cylinder.
Reduced object	Green-hyperbolic complex	metric Bach detour/BV complex	The reduced central operator need not have scalar principal symbol.
Pairing	differential pairing and Poisson structures	odd BV pairing, Green current, and residual pairing	Equality is transported up to the declared chain homotopy.
Boundaries	globally hyperbolic support categories	$\mathbb{R} \times S^3$, no timelike boundary	Boundary flux is a separate theorem.

3. Reproduced benchmark

The shared benchmark is the Green-homotopy identity and its causal difference. On the complete 386-row prolonged cylinder complex, the repository verifies coefficientwise that the retarded and advanced degree-minus-one maps satisfy the chain-homotopy equations on the declared support categories. Their difference induces the compact-to-global quasi-isomorphism, and temporal cutoffs recover the fifteen conformal-Killing endpoint classes without duplication. The direct current calculation transports the normalized H^4 pairing to I_2 .

This is the concrete analogue of the abstract causal quasi-isomorphism and Poisson-compatibility statements. The benchmark is reproduced by:

```
python3 symbolic/verify_conformal_final_covariant_transport.py
python3 symbolic/verify_conformal_covariant_H4_proof_ledger.py --check --guards
```

The exact claim and release rail are documented in [Paper 8](#) and its [reproduction guide](#).

4. Added result

Concrete transfer theorem. On the declared conformal-cylinder complex, the support-local cyclic contraction from the tractor/prolonged parent to the metric detour complex transfers retarded and advanced Green homotopies, their support properties, the causal quasi-isomorphism, and the current pairing. No canonical Green inverse of the isolated fourth-order metric operator is required.

The proof factors the construction into curvature prolongation, exact tractor-to-BGG contraction, filtered homological perturbation, finite local shears, and cyclic adjoint transport. Each factor has an explicit inverse or homotopy identity and a support ledger. Generality is currently G2: complete on one Lorentzian background. The abstract theorem with background-uniform hypotheses is not yet frozen.

5. Consequence in their language

The example supplies a large, coefficient-level fourth-order gauge complex in which Green hyperbolicity belongs to the complex even though a preferred same-bundle scalar-principal factorization of its middle operator is excluded. For detour geometry, it adds a Lorentzian propagation layer to the tractor/BGG construction. For Green-hyperbolic complexes, it suggests a practical closure criterion under cyclic differential SDRs rather than only an existence definition.

6. Scope boundary

This note does **not** establish the abstract transfer theorem on arbitrary globally hyperbolic manifolds. It does not cover timelike boundaries, interactions, a Hadamard state, time-ordered products, or the quantum master equation. The surviving H^4 classes are centered deformation classes, not one-particle gravitons. A second detour consumer is still required to separate the reusable hypotheses from cylinder-specific identities.

7. One useful question for adjacent experts

Is support-preserving **cyclic** transfer already implied by the published Green-hyperbolic-complex hypotheses when the contraction contains finite-order differential shears, or is an additional wavefront/support or pairing-compatibility hypothesis necessary? A counterexample would be as useful as a positive abstraction theorem.

Reproducibility receipt

```
source papers: arXiv:2207.04069v2; arXiv:math/0606401v2
project source: Paper 8 artifact-ready snapshot and current master
verification: commands in section 3
dependency tag: LORENTZIAN-CAUSAL
generality level: G2_COMPLETE_ONE_BACKGROUND
lifecycle state: THEOREM_EXTRACTION
claim flag: CONCRETE_CYCLIC_CAUSAL_TRANSFER_CERTIFIED
open fields: abstract hypotheses; second consumer; boundary version
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